

Giulio Segalini

📍 Neuchâtel, Switzerland ✉ giuliosegalini@protonmail.com in Giulio Segalini 🌐 gsegalini

Education

Delft University of Technology

Sept 2023 – July 2025

M.Sc. in Computer Science Cum Laude - Special 4TU program in Cybersecurity

- GPA: 8.4, Cum Laude
- **Thesis:** Fair Transaction Ordering on DAGs;
- **Relevant Coursework:** Network Security, Privacy Enhancing Technologies, Security and Cryptography, Distributed Algorithms, Cyber Data Analytics, Cyber Risk Management, Systems Security.

Delft University of Technology

Sept 2020 – June 2023

B.Sc. in Computer Science and Engineering Cum Laude

- GPA: 9.1/10, Cum Laude, Honors Program;
- **Thesis:** Optimal Decision Trees for Algorithm Selection Problem
- **Relevant Coursework:** Leadership in a Complex World (**Honors Program**), Operating Systems (**C**, **Linux**), Algebra and Cryptography, Computer Graphics (**C++**, **OpenGL**), Algorithm Design (**Java**), Computer Networks, Machine Learning.

Skills & Interests

Technologies: Cuda, Torch, Tokio, OpenGL, scrapy, Wireshark, Docker, Spring, Git, Vulkan, Latex, Blockchain, PostgreSQL, Gradle, Sklearn, RocksDB.

Interests: Capture The Flag (CTF), Hacking, Emulation, Themed Parks and Entertainment, Music, Gaming History and Retrogames, Politics, Cooking.

Programming Languages: C, C++, Rust, Python, Assembly, Scala, Haskell, Java, Agda, JavaScript, Z3.

Languages: Italian (Native), English (Fluent, main work and study language), French (Basic).

Publications

Byzantine-Resilient Federated Computation of Differentially Private Summary Statistics

ACM Middleware 2025

Giulio Segalini, Maria Fernandes, Jérémie Decouchant

[Middleware 2025](#) [🔗](#)

Experience

PhD Candidate in Ai Safety and Security

Neuchâtel, Switzerland

University of Neuchâtel

Oct 2025 - Present

- Research interests: Ai Safety and Security, Privacy, Cryptography
- Technologies and platforms used: **Python**, **Linux**, **Torch**, **CUDA**, **Docker**.

Head Teaching Assistant for Network Security, Operating Systems and Distributed Algorithms

Delft, the Netherlands

Delft University of Technology

Jan 2023 – July 2025

- Helped to organize and manage the courses on a deeper level than a normal teaching assistant. Managed labs and assistants' schedules.
- Designed and developed a standard automatic grading framework for the assignments.
- Restructured course content, created new assignments and handled grading of assignments and exams.
- Gave a lecture on Transaction Order Fairness in blockchain.
- Technologies and platforms used: **C**, **Linux**, **Docker**, **Python**, **scrapy**, **Wireshark**, **dpkt**, **ipv8**, **WebLab**.

Teaching Assistant

Delft, the Netherlands

Delft University of Technology

April 2022 – July 2025

- Helped Students understand the material, course contents and concepts. Provided support during labs.
- Graded exams and assignments.
- Courses assisted: Computer Networks, Probability Theory and Statistics, Concepts of Programming Languages, Operating Systems, Computer Graphics, Computer Security, Functional Programming, Distributed Algorithms, Network Security.

Research Assistant

Delft University of Technology

Delft, the Netherlands

July 2023 – Aug. 2023

- Worked on a research project related to training Optimal Decision Trees using Dynamic Programming.
- Technologies and platforms used: **C++**, **Python**, **R**.

Software Engineer

Video Window

Delft, the Netherlands

April 2022 – July 2022

- Designed and developed a custom **Wayland Compositor** to display content on multiple screens programmatically and with hardware acceleration support.
- Technologies and platforms used: **Rust**, **Vulkan**, **Linux**, **Wayland**.

Extracurricular Activities

Capture the Flag: Senior member of TU Delft CTF Team. Specialized in Cryptography.

- Participated and won numerous CTF hacking competitions.
- Organized TUDCTF 2023 and 2024, large events with 200+ participants.
- Best academic team in the Netherlands in 2024 and 2022, 2nd best in 2023 according to [CTFTime](#) .
- Won the academic division (2nd and 4th place overall) of GlacierCTF 2024 and 2023.
- 5th place SmileyCTF 2025
- 3rd place SunshineCTF 2024

Competitions

- European Union Science Olympiads (EUSO, now EOES) 2018 - Representative for Italy, Silver Medal;
- Italian Team Mathematics Olympiads 2019 — 4th place;
- Qualified for Benelux Algorithmic Programming Contest 2020;
- Delft First Years Programming Contest 2021 — 3rd Place.

Projects

CHIP-8 Emulator and Disassembler

[Chip=8 Emulator](#) 

- Developed a (simple) CHIP-8 emulator (interpreter) and disassembler.
- **Technologies Used:** C++.

Research on Optimal Decision Trees for Algorithm Selection

[Decision Trees](#) 

- Researched the applicability and accuracy of optimal decision trees for the algorithm selection problem.
- **Technologies Used:** C++, Java, R.

C++ Ray Tracer

- Implemented a Ray Tracer to render scenes realistically.
- Composed of numerous features, such as Depth-of-Field, Motion Blur, Mipmapping, Textures, Environment Mapping and soft shadows.
- **Technologies Used:** C++, OpenGL, GLM.